

[Final Exam (Sample)]

Spring 2024: CSE 309: Theory of Automata

Instructors: Shahid Hussain, Imran Rauf

Sunday, May 26, 2024. Total Marks: 35 (final) + 5 (quiz) + bonus, Duration: 3 hours

Name: _____, Student ID: _____

Please read these instructions carefully:

- This exam contains 8 questions. Question 1 contributes towards the Quiz component of the course. The rest of the questions contribute towards the final exam component. Question 8 is a bonus question.
- Use of calculators, mobile, or any other communication devices is strictly prohibited for the entirety of the exam period.
- Please submit your devices in your bag at the front of the examination room.
- You may keep writing material with you on your desk.
- Read all questions carefully before you start answering and prioritize which questions to attempt first.
- You may ask for extra sheets for your rough work, scribbling, future planning etc., **anything on extra sheet(s) will not be graded.**
- Please write your name and student ID on this question paper clearly.
- Please do not use pencils or red/green/yellow pens for answers that you want to be graded.
- Instructor(s) will not entertain any queries during the exam.
- Please attach all extra sheets with your answer book.
- Please write clearly and legibly and avoid overwriting.
- Pay attention to the instructions for each question.
- Acquisition of answers through unfair means will automatically cancel your exam.
- Keep track of the time.
- **Your answer(s) will not be graded if unreadable (by the instructors).**
- **Your exam will not be graded if you don't follow these instructions.**

— Good Luck —

Question 1: Quiz 5 marks

Consider the following pair of languages:

$\text{DIAG} := \{ \langle M, k \rangle : M \text{ is a square matrix and sum of values on its diagonal is at most } k \}$

$\text{TSP} := \{ \langle M, k \rangle : M \text{ is a square matrix representing inter-city distances}$

and there is tour that visit all cities of total cost at most $k \}$

Note that TSP is a known **NP**-hard problem, whereas DIAG is in **P**. Which of the following must be true, assuming $\mathbf{P} \neq \mathbf{NP}$? [No justification is needed.]

- (a) _____ $\text{TSP} \in \mathbf{P}$
- (b) _____ DIAG is in **NP**
- (c) _____ There is a polynomial-time reduction from SAT to TSP.
- (d) _____ There is a polynomial-time reduction from DIAG to TSP.

Question 2: *6 marks*

- (a) (3 marks) Fix the alphabet $\Sigma = \{0, 1\}$. A *run* in a string $w \in \Sigma^*$ is a maximal substring of consecutive identical characters. For example, the string 001110011 has exactly four runs: $001110011 = 00 \cdot 111 \cdot 00 \cdot 11$.

Let L be the set of all strings in Σ^* that contains only odd length runs. For example, L contains **000111111110** but does not contain the string **10000111**. Describe a regular expression for L .

[illegible]

- (b) (3 marks) Describe a CFG that generates the following language:

$$L = \{a^x b^y c^z : y = x + z \text{ and } x, y, z \geq 0\}$$

[illegible]

Question 3: 6 marks

Indicate whether following statements are **true** or **false** also justify your answer i.e., give a simple proof for a **true** statement and a counterexample for a **false** statement.

(a) _____ Every regular language is context-free.

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(b) _____ Following language L is **NP**-hard.

$$L = \{\langle M \rangle : M \text{ is a DFA that accepts all strings}\}$$

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Question 4: 5 marks

Show that if A is *recognizable* and $A \leq_m \overline{A}$, then A is *decidable*.

Question 5: 6 marks

Show that following problems are in class **NP**.

- (a) **3SAT:** Given a CNF formula ϕ with n variables and m clauses, is there an assignment to the variables that satisfies all the clauses.

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- (b) **Dominating Set:** Given a graph $G = (V, E)$ and a number k , is there a set $V' \subseteq V$ such that $|V'| \leq k$ and for every vertex $v \in V$, either v is in V' or v is adjacent to a vertex in V' .

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Question 6: 5 marks

If L is a language over some alphabet Σ^* then the complement of L denoted as $\bar{L}$ is defined as $\bar{L} = \Sigma^* - L$. We say a language L is in the class **coNP** if its complement $\bar{L}$ is the class **NP**. Prove/disprove that “**NP** = **coNP** if and only if $A \leq_P B$ and $B \leq_P A$ where A is an **NP**-complete languages and B is a **coNP**-complete language.”

Question 7: 7 marks

Let $T = \{\langle M \rangle \mid M \text{ is a TM that accepts } w^{\text{rev}} \text{ whenever it accepts } w\}$. Prove by reduction that T is undecidable.

Please ensure that you give a complete argument regarding why your reduction works. You can choose any undecidable language that we have seen in class for your reduction.

Question 8: Bonus *5 marks*

Show that the class **NP** is closed under the star operation. That if L is in **NP** then L^* is also in **NP**.